Capstone hypothesis and baseline

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Capstone hypothesis and baseline before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Documenting a network experiment

Without opening the old lesson, explain one key idea from Documenting a network experiment. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Dataset shift in neural models

Without opening the old lesson, explain one key idea from Dataset shift in neural models. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Similarity and normalisation

Without opening the old lesson, explain one key idea from Similarity and normalisation. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Lesson reference for workbook practice

Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

A capstone needs a question that **evidence** can answer. “Build a better network” leaves better undefined.

Here the question is: does two-update full-batch **fitting** of one scalar ReLU weight reduce validation mean squared error relative to the initial-weight **baseline** on a constructed positive-**input** task?

Predict max(0, a*x), with no bias. The **baseline** has a = 1. The trained candidate starts at a = 1, uses learning rate 0.1 and exactly two updates. These choices are written before results are inspected.

The **hypothesis** is that the trained candidate has lower validation MSE. The plan does not guarantee that result. This task tests **training** and record keeping, not whether neural networks are preferable to every other model.

A stronger comparison may also include an ordinary fitted line.

WORKBOOK · 2 OF 15

Practise with fading help

> Complete the supported Capstone hypothesis and baseline trace and record the first step you can explain without help.

Use W2: Calculate the reference score for Capstone **hypothesis** and **baseline**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

WORKBOOK · 3 OF 15

Individual reasoning check

> Check the proposed Capstone hypothesis and baseline explanation against the supplied facts before discussing it.

Use the worked case for Capstone **hypothesis** and **baseline**. Proposed explanation: “A **hypothesis** is a result”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Plan a new case independently

A new task predicts max(0, a*x), with **baseline** a = 1. **Training** rows are (1,3) and (2,6); validation rows are (3,9) and (4,12); final rows are sealed.

Specify a candidate starting at 1 with learning rate 0.1 and two full-batch updates. Calculate **baseline** validation MSE and write a strict improvement criterion. Has the candidate succeeded yet?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Write the hypothesis

> Write the hypothesis. Apply the supplied Capstone hypothesis and baseline facts and keep your reasoning.

State the **prediction** rule, **baseline**, candidate learning rate and update count, comparison data and metric. Write a testable **hypothesis**.

My reasoning and evidence

WORKBOOK · 6 OF 15

W2 Calculate the reference score

> Calculate the reference score. Apply the supplied Capstone hypothesis and baseline facts and keep your reasoning.

Compute **baseline prediction**s, errors, squared errors and MSE on V1 and V2. State the denominator.

My reasoning and evidence

WORKBOOK · 7 OF 15

W3 Audit the plan

> Audit the plan. Apply the supplied Capstone hypothesis and baseline facts and keep your reasoning.

A proposal says “try models until the test score looks good”. Identify the missing fixed choices and explain why that test is not independent.

My reasoning and evidence

WORKBOOK · 8 OF 15

W4 Plan a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Plan a new case independently. Apply the supplied Capstone hypothesis and baseline facts and keep your reasoning.

A new task predicts max(0, a*x), with **baseline** a = 1. **Training** rows are (1,3) and (2,6); validation rows are (3,9) and (4,12); final rows are sealed.

Specify a candidate starting at 1 with learning rate 0.1 and two full-batch updates. Calculate **baseline** validation MSE and write a strict improvement criterion. Has the candidate succeeded yet?

My reasoning and evidence

WORKBOOK · 9 OF 15

W5 Preserve the reserved evidence

> Preserve the reserved evidence. Apply the supplied Capstone hypothesis and baseline facts and keep your reasoning.

A partner inspects final labels and changes the learning rate. Explain how to document this and obtain independent **evidence**.

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Capstone hypothesis and baseline to Image-pattern Network Investigation, identify a limit and choose the next useful test.

For Image-pattern Network Investigation, use this week's independent task as a rehearsal for the final **evidence** pack: A new task predicts max(0, a*x), with **baseline** a = 1.

Training rows are (1,3) and (2,6); validation rows are (3,9) and (4,12); final rows are sealed. Specify a candidate starting at 1 with learning rate 0.1 and two full-batch updates.

Calculate **baseline** validation MSE and write a strict improvement criterion. Has the candidate succeeded yet?

Keep actual **observation**s separate from expected results, preserve failures, compare a **baseline** and explain the limits.

> **My project change, supporting evidence and remaining limit**

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	

Evidence	My record
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 12 OF 15

Pictorial case practice: Capstone hypothesis and baseline

> A testable image-pattern hypothesis. A team proposes that including the centre-pixel feature improves a small fixed network's classification of toy image patterns. Explain the labelled step before checking the changed condition.

A testable image-pattern **hypothesis**. A team proposes that including the centre-pixel feature improves a small fixed network's classification of toy image patterns.

Two frozen versions are planned: A uses corner features only; B also uses the centre feature. Eight reserved toy patterns have labels defined before scoring.

AI ACADEMY / AI-5/33

Capstone hypothesis and baseline

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Write the prediction

B should have more correct labels than A on the same eight patterns.

2. Declare the baseline

A is a documented corner-only network, not a deliberately broken version.

3. Reserve evaluation

Keep the eight patterns out of model selection and tuning.

CHANGE ONE THING

Give only B the eight evaluation labels during training.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Name the intended change, the **baseline** and the measured outcome. Why must both versions avoid **training** on the eight reserved labels?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Image-pattern Network Investigation, save the labelled trace and the changed-case explanation beside the experiment record.

State exactly what this example supports before linking it to the wider capstone claim.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Image-pattern Network Investigation record.

Project: Image-pattern Network Investigation

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

For the Image-pattern Network Investigation, save the labelled trace and the changed-case explanation beside the experiment record. State exactly what this example supports before linking it to the wider capstone claim.

Evidence to keep	My record
This week's input and deciding step	

Evidence to keep	My record
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A testable image-pattern **hypothesis**. A team proposes that including the centre-pixel feature improves a small fixed network's classification of toy image patterns.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Keep the eight patterns out of model selection and tuning.

1. Reserve evaluation

2. Declare the **baseline**

3. Write the **prediction**

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Reserve evaluation Keep the eight patterns out of model selection and tuning. A capstone **hypothesis** links a specific architectural change to a measured outcome.

A **baseline** anchors the comparison; frozen labels and held-out patterns protect it from answer-key tuning.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Give only B the eight evaluation labels during **training**. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. The proposed comparison becomes contaminated rather than a valid held-out test.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The proposed comparison becomes contaminated rather than a valid held-out test. One version has gained privileged information from the reserved evaluation set.

Repair a missing building block

Review Documenting a network experiment (ai-5/32). Goal: Record a network experiment with enough detail to replay its calculations. Short practice: What additional **evidence** is needed for a final performance claim?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Dataset shift in neural models (ai-5/30). Goal: Explain how a changed **input** mix can change overall accuracy. Short practice: Did the supplied model's dim-light accuracy deteriorate?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Similarity and normalisation (ai-5/27). Goal: Calculate vector length and normalise a nonzero vector. Short practice: Why do the original two metrics select different winners?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
>
> | | |
> |---|---|
> | | |
> | | |
> | | |

WORKBOOK · 15 OF 15

Visual concept journey

> Explain each Capstone hypothesis and baseline picture from its labels, then record a relevant limit.

01. Capstone hypothesis and baseline: the case

AI ACADEMY / AI-5/33 · 01 · LOOK AT THE FACTS

Capstone hypothesis and baseline: the case

The situation

A testable image-pattern hypothesis. A team proposes that including the centre-pixel feature improves a small fixed network's classification of toy image patterns.

Supplied evidence

Two frozen versions are planned: A uses corner features only; B also uses the centre feature. Eight reserved toy patterns have labels defined before scoring.

Explain capstone hypothesis and baseline with a traceable toy-network case and a bounded conclusion.

Follow the labelled picture with the matching explanation. Open the vector diagram

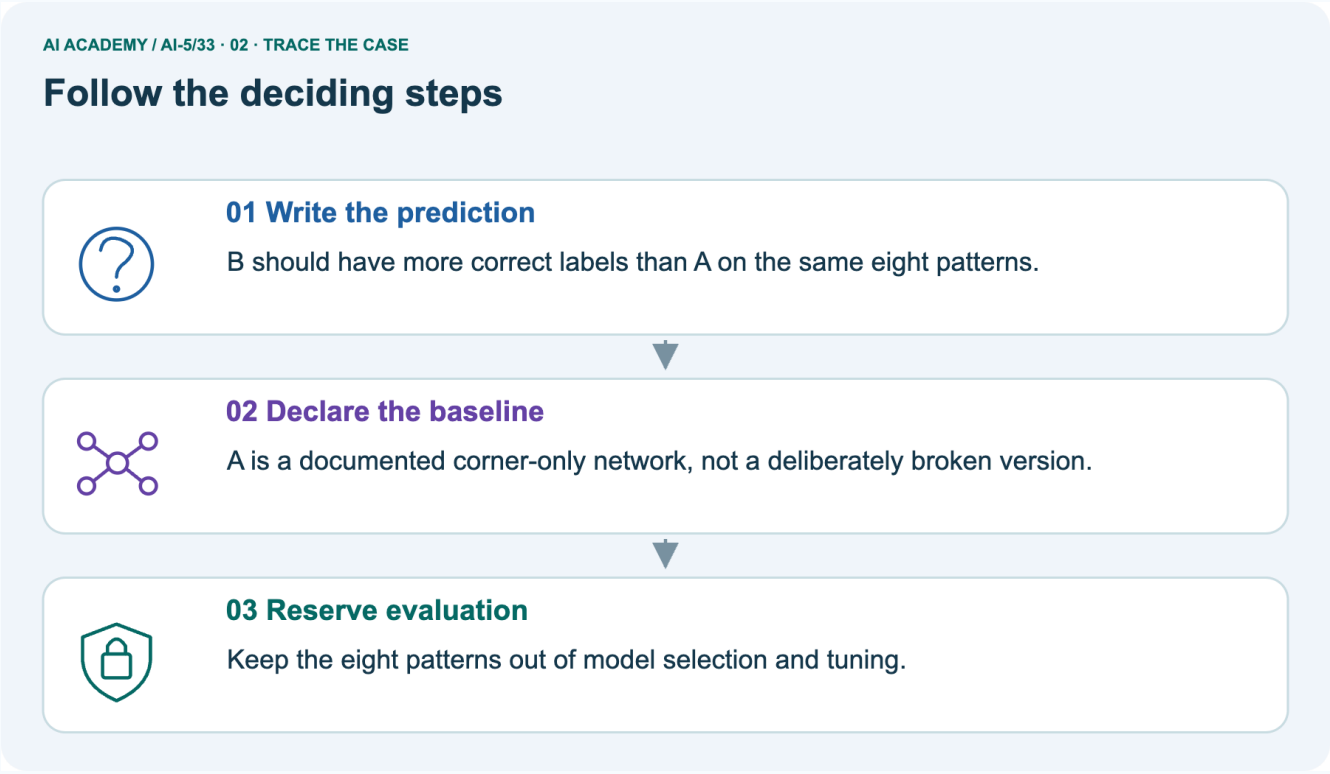
A testable image-pattern **hypothesis**. A team proposes that including the centre-pixel feature improves a small fixed network's classification of toy image patterns.

Supplied facts: Two frozen versions are planned: A uses corner features only; B also uses the centre feature. Eight reserved toy patterns have labels defined before scoring.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



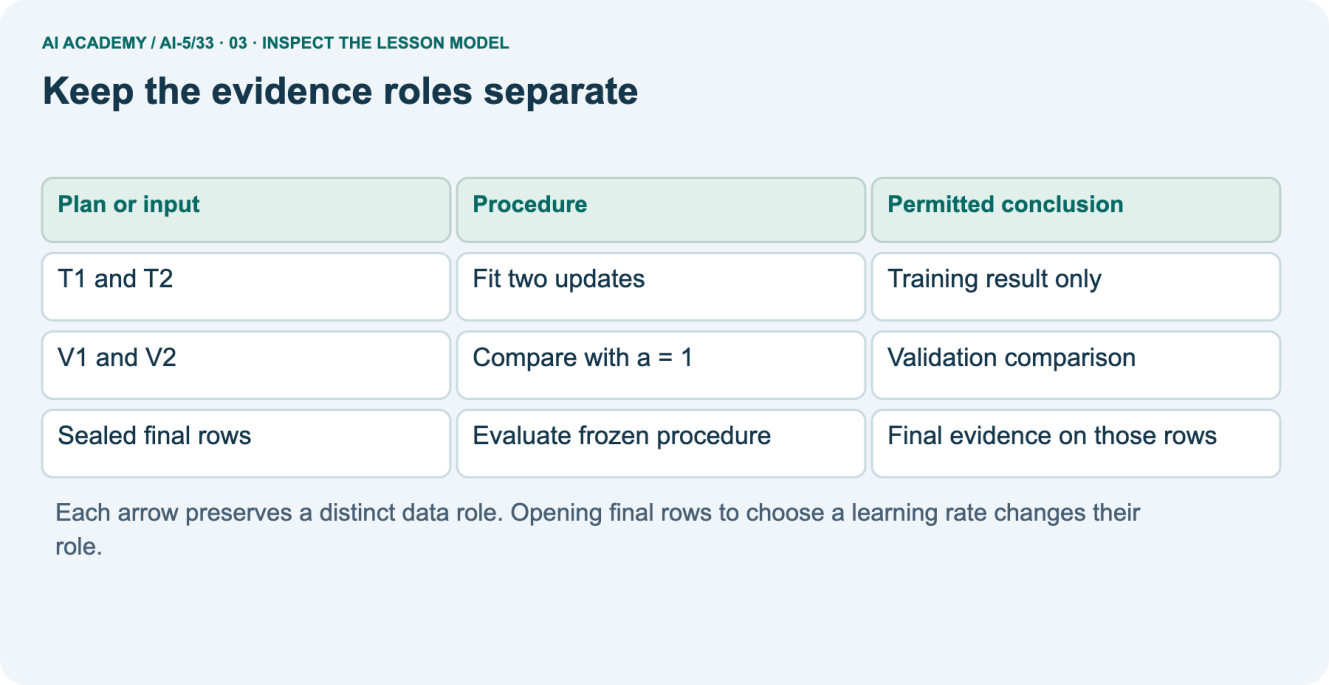
Follow the labelled picture with the matching explanation. Open the vector diagram

Write the **prediction**: B should have more correct labels than A on the same eight patterns. → Declare the **baseline**: A is a documented corner-only network, not a deliberately broken version. → Reserve evaluation: Keep the eight patterns out of model selection and tuning.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Keep the evidence roles separate



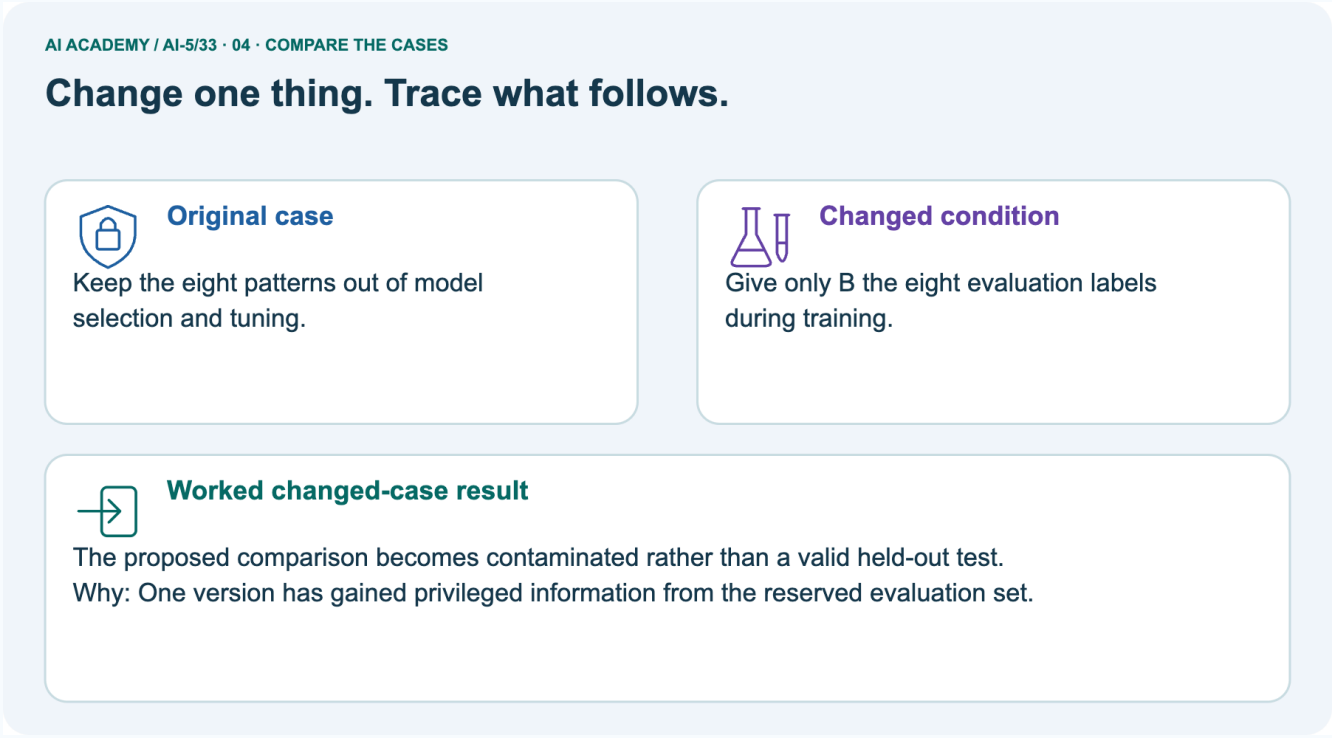
Follow the labelled picture with the matching explanation. Open the vector diagram

Keep the **evidence** roles separate. T1 and T2 | Fit two updates | **Training** result only. V1 and V2 | Compare with a = 1 | Validation comparison. Sealed final rows | Evaluate frozen procedure | Final **evidence** on those rows

> Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

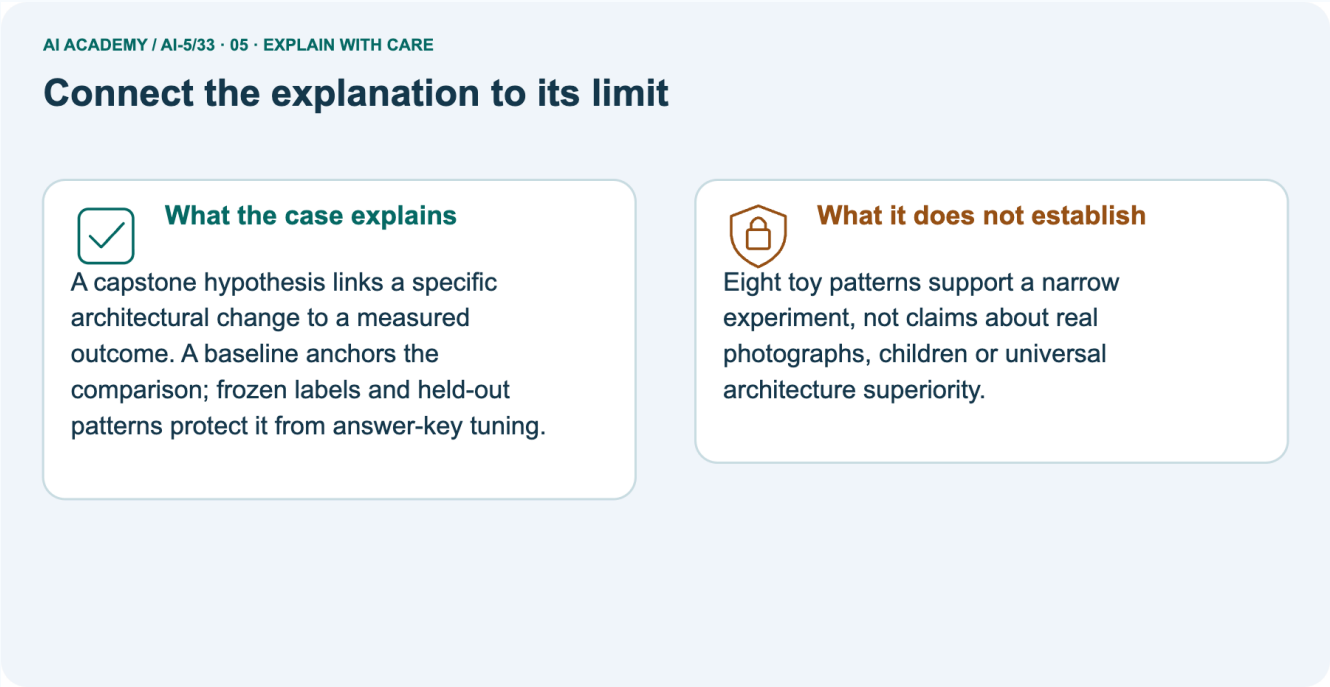
Original case: Keep the eight patterns out of model selection and tuning. Changed condition: Give only B the eight evaluation labels during **training**.

Expected result: The proposed comparison becomes contaminated rather than a valid held-out test. Reason: One version has gained privileged information from the reserved evaluation set.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

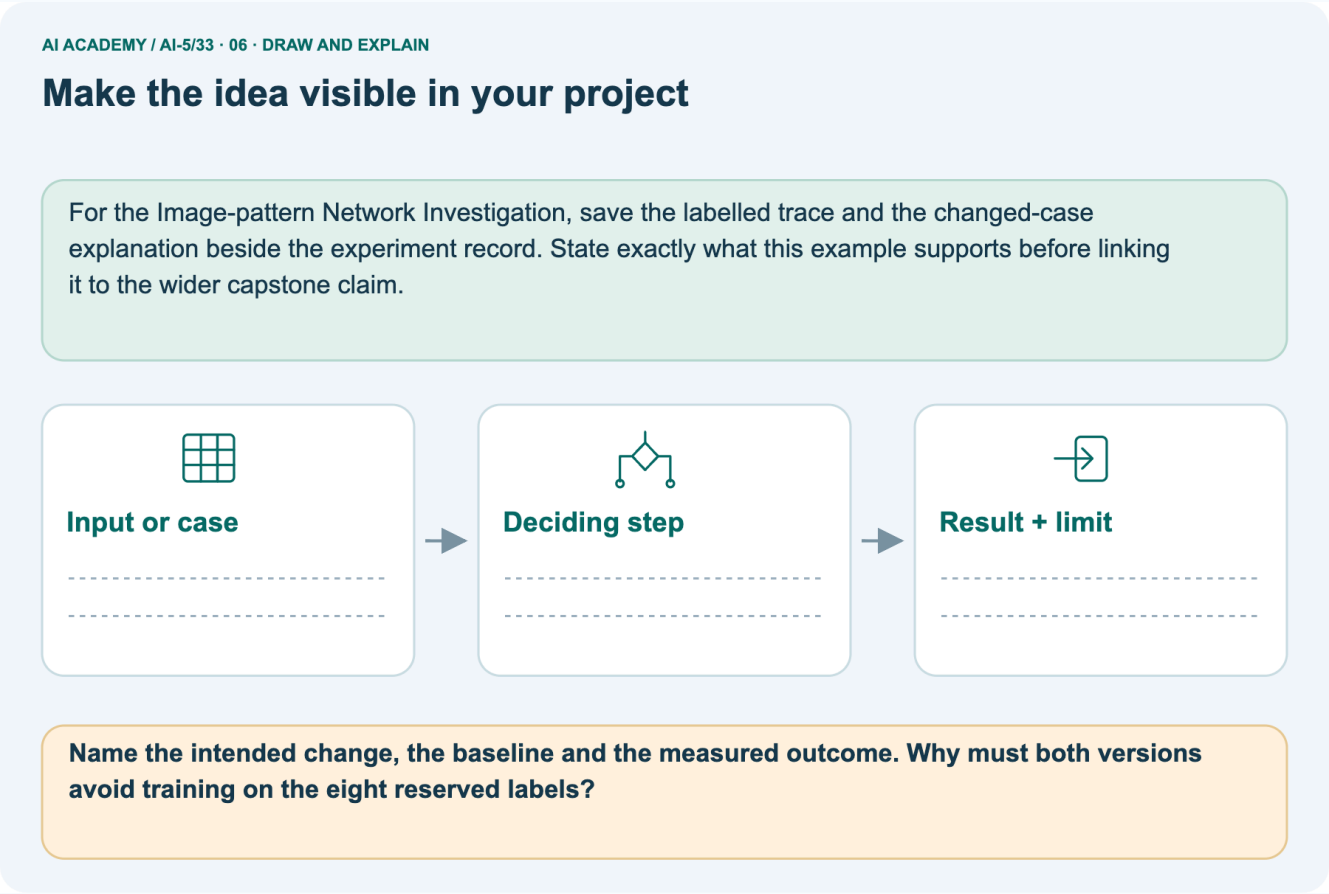
A capstone **hypothesis** links a specific architectural change to a measured outcome. A **baseline** anchors the comparison; frozen labels and held-out patterns protect it from answer-key tuning.

Limit: Eight toy patterns support a narrow experiment, not claims about real photographs, children or universal architecture superiority.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Image-pattern Network Investigation, save the labelled trace and the changed-case explanation beside the experiment record. State exactly what this example supports before linking it to the wider capstone claim.

Draw your **input**, deciding step, result and limit. Supported question: Name the intended change, the **baseline** and the measured outcome. Why must both versions avoid **training** on the eight reserved labels?

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.